

A Fresh Perspective on the Collatz Problem and Generalized Algorithms

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May 25–27, 2025

Preface

And just as a disclaimer, the try is completely innovative and bold, so please before you start judging please change your lenses ;). Thanks in advance my dears!

Also note: I AM NOT A PROFESSIONAL MATHEMATICIAN AND THAT THIS IS ONLY AMATEUR TEENAGE WORK DONE WITH VERY LITTLE RESSOURCES, anyway sending LOVE!

Abstract

This is in some kind a simple and innocent try to tackle one of the craziest conjectures by a random kid out in this world, only with a pen, papers – a lot –, and maybe even an old Nokia phone – RIP – for instant research (it had data, don't worry;)). YEAH! Life's easy, anyway. Here we go again..

Introduction

The Collatz conjecture states that if you take any natural number, and see if it's even then divide it by 2, and if it's odd multiply it by 3 then add 1. And in order to prove it, we will study the algorithm from a pretty new point of view: when it reaches the end goal (1).

Try

Let the generalized algorithm:

$$\begin{cases} \text{If } x \mid n, \text{ then } n/x \\ \text{If not, } kn + \frac{n+\alpha}{x} \quad \text{or} \quad kn + \frac{n-\alpha}{x} \end{cases}$$

all in $\mathbb{N}^*$ and α is the smallest number to either add or subtract from n to reach x or a multiple of it.

Note that our goal is for any n (a variable), the algorithm always reaches 1, according to k, x , and α (well defined unknowns).

And in order to achieve that goal, at some point we should have:

$$n \pm \alpha = x \quad \text{or} \quad n = x$$

Because we need to have at some point:

$$\frac{n}{x} = 1 \quad \text{or} \quad \frac{n \pm \alpha}{x} = 1$$

to eventually reach 1.

Why did we neglect the "kn"?

We did because the algorithm should always reach 1, and since with INTUITION the algorithm:

$$\begin{cases} \text{If } x \mid n, \text{ then } n/x \\ \text{If not, } \frac{n+\alpha}{x} \quad \text{or} \quad \frac{n-\alpha}{x} \end{cases}$$

all in $\mathbb{N}^*$ always reaches 1!

And in order to achieve our goal (again), our algorithm should enter the following series:

$$x^2, x, x^0 \quad \text{which is in other words: } x^2, x, 1.$$

Why?

Simply, to reach 1 with $n \neq 1$ it should reach $n = x$ and the closest step is $n = x^2$.

We will MAYBE analyse the general case $n = x^y$ at the end of the paper. Stay tuned.

Anyway, we got two cases (except the easy one: numbers that only need to be divided by x to reach 1).

Case 1:

$$kn + \frac{n \pm \alpha}{x} = \frac{x^2}{x} \quad (\text{because we are dividing after by } x)$$

$$kn + 1 = x$$

$$kn = x - 1$$

$$k = \frac{x - 1}{n}$$

$$k = \frac{n \pm \alpha - 1}{n}$$

$$k = 1 + \frac{\pm \alpha - 1}{n}$$

And since $\alpha < n$, to reach 1 and not diverge, and we need to have $k \in \mathbb{N}^*$

We got to have $\pm \alpha = 1$ from which we have $k = 1$ and if not we will have $k = 1 + z$ with $z \notin \mathbb{N}$ but $z \in \mathbb{Q}$.

So for the algorithm to work we should have:

$$\begin{cases} k = 1 \\ \alpha = \pm 1 \end{cases}$$

Case 2:

$$kn + \frac{n \pm \alpha}{x} = \frac{x}{x} \quad (\text{because we are dividing after by } x)$$

$$kn + 1 = 1$$

$$kn = 0$$

And since n is a variable that never reaches 0, because the algorithm never allows it, which means that $k = 0$.

Interpretation

The first case shows us that from an intuitive algorithm ($k = 0$) we can get another one giving the same result (1) with

$$\begin{cases} k = 1 & (\text{independently from any case}) \\ \alpha = \pm 1 & (\text{according to the specific case}) \end{cases}$$

however it should enter the famous series: $x^2, x, 1$ or $x, 1$, and why is it just these? Take the smallest number that isn't a multiple of x , and for the $x = 2$ case and do the $3n + 1$ with $n = 1$, and you get 4, same thing $x = 3$ but in 2 case, with the two smallest numbers not multiples of 3: 2 and 1, and you get 9 and 3 respectively, see?

And the second case is just the intuitive algorithm, but, it shows us that it doesn't need the " $x^2, x, 1$ " series however it only needs the " $x, 1$ " one, and that isn't monitored by the number α , because it doesn't appear when trying to solve the algorithm's unknowns so it is just as n , a variable that changes according to the specific case. WHICH MEANS IN OTHER WORDS THAT THE INTUITIVE ALGORITHM WORKS FOR ANY x , unlike the first case..

Note

The first case algorithm doesn't work for any x , because with

$$\begin{cases} k = 1 \\ \alpha = \pm 1 \end{cases}$$

4 isn't always reachable nor any of its multiples, from which we generalize for $x \geq 4$.

So the lucky values of x are 2 and 3 (Collatz & Syracuse).

NOW THE QUESTION IS: CAN WE REACH A GENERALIZED FORM FOR ANY x ?

The proof is left as a challenge for the reader, if you do it contact me in this life or in the afterlife.

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I don't think so, because I think I found it my dear friends, and it is as follows:

Here you go:

$$\begin{cases} \text{If } x \mid n, \text{ then } n/x \\ \text{If not, } n + \frac{n+\alpha}{x} \end{cases}$$

all in $\mathbb{N}^*$ and α is the smallest number to either add or subtract from n to reach x or a multiple of it.

Let $\beta = x \bmod n$

And if $\beta \leq n/2$ then $\alpha = -\beta < 0$

Or if $\beta \geq n/2$ then $\alpha = n - \beta > 0$

And if you ever stumble upon a loop or start to diverge then change the formula, just a little advice ;).

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